NATIONAL ECONOMICS UNIVERSITY, HANOI
COLLEGE OF TECHNOLOGY
FACULTY OF DATA SCIENCE AND ARTIFICIAL INTELLIGENCE

# TIME-SERIES ANALYSIS AND FORECASTING

---

## REPORT

# Topic: Human Vital Signs

---

Instructor: Nguyen Trong Nghia
Student: Phan Hai Dang 11247270

Hanoi, Month 5 Year 2026

# Contents

# 1 Introduction

In modern healthcare systems, vital signs are among the first clinical indicators used to assess a patient's condition. When a patient is admitted to a hospital or examined in an emergency setting, measurements such as heart rate, respiratory rate, body temperature, blood pressure, and oxygen saturation provide immediate insight into their physiological status. These indicators help medical professionals quickly determine whether a patient is stable, requires close monitoring, or may be at risk of serious complications.

With the rapid growth of wearable devices, remote patient monitoring systems, and digital healthcare technologies, the collection of physiological data has become increasingly continuous and automated. This creates valuable opportunities for applying time series analysis to identify patterns, detect abnormalities, and forecast future physiological changes.

Forecasting human vital signs is particularly important because unexpected deviations may indicate early signs of health deterioration. For example, a sudden increase in heart rate or respiratory rate may signal cardiovascular stress, infection, or respiratory distress. Accurate prediction models can therefore support proactive clinical decision-making and improve patient safety.

This project investigates the temporal behavior of physiological measurements using the **Human Vital Signs Dataset**. Since these measurements are recorded sequentially, they naturally exhibit time-dependent structures such as trends, autocorrelation, and seasonal fluctuations, making them suitable for time series modeling.

The primary objective of this study is to analyze these temporal patterns and evaluate forecasting performance using classical statistical time series models, specifically:

- AutoRegressive Integrated Moving Average (ARIMA)
- Seasonal AutoRegressive Integrated Moving Average (SARIMA)

The performance of these models is assessed using standard evaluation metrics, including:

- Root Mean Squared Error (RMSE)
- Mean Absolute Error (MAE)
- Akaike Information Criterion (AIC)

To extend the analysis beyond forecasting, an additional machine learning experiment is conducted using a Random Forest classifier to predict patient risk categories based on physiological measurements. This extension allows further exploration of how vital sign patterns relate to overall patient risk assessment.

# 2 Methodology

This section describes the data preprocessing procedures, time series modeling techniques, evaluation metrics, and machine learning extension used in this project.

## 2.1 Dataset Description

The dataset used in this study is the **Human Vital Signs Dataset**, which contains physiological measurements collected from multiple patients over time.
The dataset includes several important health indicators:

- Heart Rate
- Respiratory Rate
- Body Temperature
- Oxygen Saturation
- Systolic Blood Pressure
- Diastolic Blood Pressure
- Weight
- Height
- Derived Heart Rate Variability
- Risk Category

These variables represent essential indicators of cardiovascular and respiratory health.
Several derived variables were also included in the original dataset, such as:

- Derived Pulse Pressure
- Derived Body Mass Index (BMI)
- Derived Mean Arterial Pressure (MAP)

These features were initially retained during exploratory analysis but later excluded from Random Forest classification to avoid information leakage, since they are directly calculated from existing measurements.

## 2.2 Data Preprocessing

Raw physiological data often contains inconsistencies, missing values, and irregular temporal spacing. Therefore, preprocessing was required before modeling.
The preprocessing pipeline consisted of the following steps:

1. Removing missing observations
2. Sorting records chronologically
3. Converting timestamps into proper datetime format

4. Aggregating measurements into hourly intervals
5. Preparing lagged features for machine learning models

The implementation workflow is illustrated below.

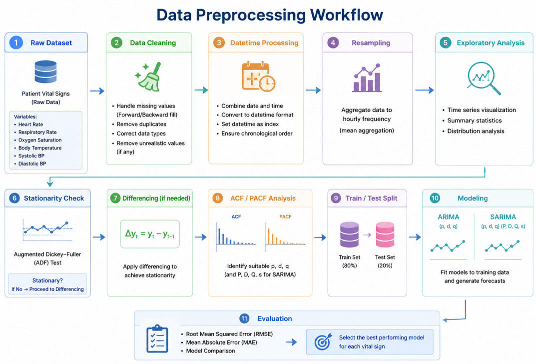


Figure 1: Data preprocessing workflow

Hourly aggregation was chosen to reduce short-term noise while preserving meaningful temporal trends.

## 2.3 Exploratory Time Series Analysis

Before fitting forecasting models, exploratory analysis was conducted to identify statistical properties of the time series.
This stage included:

- Trend visualization
- Distribution analysis
- Rolling mean and rolling variance inspection
- Detection of temporal dependencies

Visual inspection provided initial evidence of autocorrelation and possible seasonality.

## 2.4 Stationarity Analysis

Stationarity is a critical assumption for ARIMA-based forecasting.
A stationary time series has constant statistical properties over time, including mean and variance.
To evaluate stationarity, the following methods were used:

- Rolling statistics visualization
- Autocorrelation Function (ACF)
- Partial Autocorrelation Function (PACF)

When non-stationarity was detected, differencing was applied to stabilize the series.

## 2.5 ACF and PACF Analysis

ACF and PACF plots were used to estimate appropriate ARIMA parameters.
**ACF** measures correlation between observations across different lags.
**PACF** measures direct correlation after controlling for intermediate lags.
These plots provided guidance for selecting:

- Autoregressive order ($p$)
- Differencing order ($d$)
- Moving average order ($q$)

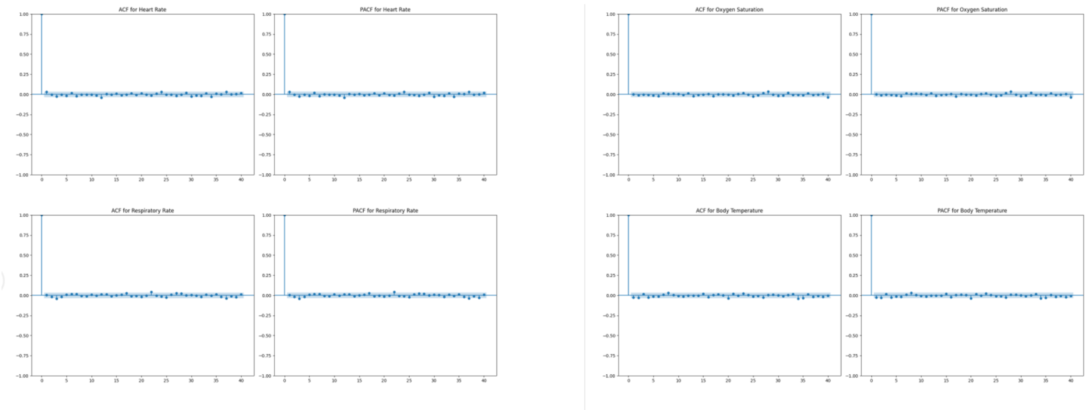

Figure 2: ACF and PACF plots

## 2.6 ARIMA Modeling

The first forecasting approach applied was the AutoRegressive Integrated Moving Average model.
The general ARIMA formulation is:

$$ARIMA(p, d, q)$$

where:

- $p$: autoregressive order
- $d$: differencing order
- $q$: moving average order

ARIMA captures linear temporal dependencies by combining autoregression, differencing, and moving average components.
Model parameters were selected using Akaike Information Criterion (AIC).

## 2.7 SARIMA Modeling

Since physiological measurements may contain periodic fluctuations, a Seasonal ARIMA model was also implemented.
The SARIMA formulation is:

$$SARIMA(p, d, q)(P, D, Q)_m$$

where:

- $(P, D, Q)$ are seasonal parameters
- $m$ is the seasonal period

Parameter optimization was performed using the `auto_arima()` procedure.
This method automatically selects the parameter combination with the lowest AIC.

## 2.8 Model Evaluation Metrics

Forecasting accuracy was evaluated using three metrics.

## 2.9 Model Evaluation Metrics

To evaluate forecasting performance, three standard metrics were used: Akaike Information Criterion (AIC), Root Mean Squared Error (RMSE), and Mean Absolute Error (MAE).
**Akaike Information Criterion (AIC)** is used to compare model quality while considering model complexity. Lower AIC values indicate a better balance between goodness-of-fit and simplicity.
**Root Mean Squared Error (RMSE)** measures the average magnitude of prediction errors, with larger errors receiving greater penalties. This metric is useful for evaluating forecasting stability.
**Mean Absolute Error (MAE)** calculates the average absolute difference between predicted and actual values. It provides an intuitive interpretation of average forecasting error.
In this study, AIC was primarily used for model selection, while RMSE and MAE were used to assess forecasting accuracy. Models with lower values across these metrics were considered to have better overall performance.

# 3 Results and Discussion

## 3.1 Model Performance Comparison

The forecasting performance of ARIMA and SARIMA models was evaluated using Root Mean Square Error (RMSE) and Mean Absolute Error (MAE) across six vital sign variables, including Heart Rate, Respiratory Rate, Oxygen Saturation, Body Temperature, Systolic Blood Pressure, and Diastolic Blood Pressure.
The experimental results indicate that both ARIMA and SARIMA produced nearly identical forecasting performance for all monitored physiological signals. This suggests that the dataset does not exhibit strong seasonal characteristics within the selected seasonal periodicity.
Among all variables, Body Temperature achieved the lowest forecasting error, indicating high temporal stability and predictability. Oxygen Saturation and Respiratory Rate also demonstrated relatively low prediction errors, suggesting smooth short-term dynamics.
In contrast, Heart Rate and Blood Pressure measurements presented comparatively higher forecasting errors, reflecting greater variability and more complex temporal fluctuations. These physiological indicators are typically more sensitive to rapid changes in patient condition, making them inherently more difficult to forecast accurately.

## 3.2 ARIMA versus SARIMA Analysis

Although SARIMA extends ARIMA by incorporating seasonal components, no significant improvement was observed in this study.
The identical error metrics obtained from both models imply that:

- The physiological time series do not contain clear seasonal patterns;
- The selected seasonal period was not strongly representative of periodic behavior;
- Non-seasonal autoregressive dynamics dominate the dataset.

Therefore, the simpler ARIMA model is considered more appropriate for this dataset due to its lower computational complexity and equivalent predictive performance.

## 3.3 Interpretation of Findings

The findings suggest that short-term human vital sign measurements are primarily influenced by immediate physiological fluctuations rather than long-term repeating cycles.
This observation aligns with real-world clinical monitoring scenarios, where sudden changes in heart rate or blood pressure are often caused by transient biological responses rather than regular periodic patterns.
Consequently, linear time series models such as ARIMA are effective for short-horizon forecasting of patient vital signs.

Figure 3: logo NEU

# 4 Discuss

Although the forecasting results were satisfactory, several limitations remain.
First, the dataset lacked strong seasonal characteristics, reducing the effectiveness of seasonal modeling approaches.
Second, the study focused on univariate forecasting, without considering interactions between multiple physiological variables.
Future work could investigate:

- Multivariate time series forecasting
- Deep learning models such as LSTM
- Real-time physiological monitoring systems
- Integration of external contextual variables

These directions may further improve predictive accuracy and practical applicability in healthcare monitoring systems.